

PROTOTYPE-DEMO(20%-MATHEMATICAL IMPLEMENTATION)

IMPLEMENTED THE MATHEMATICAL TERS LIKE DENSITY MATRIX,PAULI VECTOR.



quantum-qds-prototype

```

├── .gitignore
└── backend
    ├── detection
    │   └── init.py
    └── quantum
        ├── __init__.py
        ├── disturbance.py
        ├── fidelity.py
        ├── geometry.py
        ├── measurement.py
        ├── pauli_measurement.py
        ├── shot_experiment.py
        ├── state.py
        ├── statistical_detector.py
        └── test_state.py
  
```

```

hari-harasa-sudhan-s@Oppenheimer: ~/quantum-qds-prototype/backend/quantum
~/quantum-qds-prototype/backend/quantum
hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype$ cd ~/quantum-qds-prototype
hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype$ source backend/venv/bin/activate
(venv) hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype$ git status
On branch master
nothing to commit, working tree clean
(venv) hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype$ git log --oneline
b326768 (HEAD -> master) v0.1: quantum state analysis foundation
(venv) hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype$ cd backend/quantum
(venv) hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype/backend/quantum$ python test_state.py

Quantum State:
[0.70710678+0.j 0.70710678+0.j]

Density Matrix:
[[0.5+0.j 0.5+0.j]
 [0.5+0.j 0.5+0.j]]

Pauli Expectation Vector:
[1. 0. 0.]

Bloch Vector Norm:
1.0000000000000002

Quantum Fingerprint:
[ 1.0000000e+00  0.0000000e+00  0.0000000e+00 -4.4408921e-16
  1.0000000e+00  1.0000000e+00  1.0000000e+00]
(venv) hari-harasa-sudhan-s@Oppenheimer:~/quantum-qds-prototype/backend/quantum$ cd ~/quantum-qds-prototype
source backend/venv/bin/activate
cd backend/quantum
  
```